Yiğit Narter

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Education

University of Michigan — *Ann Arbor, MI, USA*
Ph.D. in Electrical and Computer Engineering — *Sept 2025 – Present*
GPA: 4.00/4.00 · Advisor: Prof. Inigo Incer
Relevant coursework: Nonlinear Systems, Linear Systems Theory, System-Level Analysis and Design, Linear Programming

Bilkent University — *Ankara, Türkiye*
B.S. in Electrical and Electronics Engineering — *Sept 2021 – June 2025*
GPA: 3.99/4.00 · Ranked 1st of 145 · High Honors
Relevant coursework: Feedback Control Systems, Signals and Systems, Probability and Statistics, Digital Communications, Engineering Mathematics

École Polytechnique Fédérale de Lausanne (EPFL) — *Lausanne, Switzerland*
Exchange Student in Electrical Engineering · SEMP Scholar — *Feb 2024 – July 2024*
Grade: 5.33/6.00
Relevant coursework: Machine Learning, Computer Vision, Signal Processing for Communications

Publications & Preprints

Yiğit Narter, Emily Zhao, and Inigo Incer. "Computing GR(1) Assume-Guarantee Contracts." *arXiv preprint*, 2026; *submitted to TACAS*. [arXiv]

Yiğit Narter and Inigo Incer. "On a Gradation for Asymptotic Stability." *arXiv preprint*, 2026; *under review at ACC*. [arXiv]

Yiğit Narter, Alihan Hüyük, Mihaela van der Schaar, and Cem Tekin. "Unified Screening for Multiple Diseases." *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, **PMLR 267:45737–45769**, 2025. [PMLR] [Code] [Presentation]

Research Experience

Ph.D. Researcher — *Ann Arbor, MI, USA*
Complex Engineering Systems Lab, University of Michigan — *Aug 2025 – Present*
Advisor: Prof. Inigo Incer

- Develop mathematical and computational methods for **compositional design and verification**, combining formal specifications, optimization, and reasoning under uncertainty.
- Developed theory and algorithms for **temporal/GR(1) assume–guarantee contracts**, including context-aided variable elimination, composition, and quotient; implemented the framework in **Pacti** using Python and Z3. *Preprint submitted to TACAS.*
- Developing **probabilistic assume–guarantee contracts** with linear constraints on event probabilities; using **linear algebra, LP duality, Farkas certificates, and polyhedral projection** to compute contract operations without assuming independent component behaviors. *Manuscript in preparation.*
- Introduced **degree-m stability** to characterize algebraic convergence rates of nonlinear systems, with direct and converse **Lyapunov conditions**. *Preprint under review at ACC.*

Undergraduate Researcher — *Ankara, Türkiye*
CYBORG Lab, Bilkent University — *Aug 2024 – June 2025*
Advisor: Prof. Cem Tekin

- Developed a probabilistic framework for **multi-disease screening under competing risks**, accounting for screening budgets and diagnostic-error constraints; first-author paper at **ICML 2025**.
- Formulated optimal screening/referral decisions as a **linear program** and used **Lagrangian duality** to derive risk-dependent optimal policy structure; evaluated policies using Python, CVXPY, JAX, and Monte Carlo simulation.

Undergraduate Researcher — *Lausanne, Switzerland*
Computer Vision Laboratory (CVLab), EPFL — *Jan 2024 – July 2024*

- Developed and integrated a **deep-learning-based 3D neuron segmentation pipeline** into Vaa3D and napari using Python and C++.
- Trained and evaluated **CNN/U-Net architectures** with unsupervised transfer learning across heterogeneous neuron-imaging datasets.

Industry Experience

Software Engineering Intern *Ankara, Türkiye*
HAVELSAN *Aug 2024 – Sept 2024*

- Developed a **real-time multi-camera panorama-stitching system** for augmented-reality applications using Python and OpenCV.
- Designed a calibration pipeline using SURF, RANSAC, and homography estimation, reducing online computation by precomputing inter-camera transformations.

Radar Engineering Intern *Ankara, Türkiye*
METEKSAN Defence *Aug 2023*

- Developed a radar-altimeter simulator with a Qt GUI and deployed the simulation to a **Tiva C embedded platform**, implementing UART communication and DMA-based data transfer.

Honors and Awards

Academic Excellence Award, *Bilkent University* *June 2025*
Awarded to graduating students with a cumulative GPA above 3.90/4.00.

Research Excellence Award, *Bilkent University* *June 2025*
Awarded for outstanding undergraduate research achievements, including research publications.

Swiss-European Mobility Programme (SEMP) Scholarship, *EPFL* *2024*

Comprehensive Scholarship, *Bilkent University* *2021 – 2025*
Full tuition scholarship with monthly stipend.

Ranked 56th among 2.6 million candidates, *Turkish National University Entrance Examination (YKS)* *Aug 2021*

Selected Projects

Nonlinear Control and Observer Design for an Inverted Pendulum *Winter 2026*

- Modeled and linearized a nonlinear 4-state inverted-pendulum-on-cart system; analyzed controllability and observability and designed stabilizing state-feedback control via pole placement.
- Designed and compared Luenberger and nonlinear state observers, evaluating estimation error, transient response, and closed-loop stability through nonlinear MATLAB simulations.
- Developed a nonlinear observer that reduced estimation error and increased the empirically tested stable initial-angle range from **0.6316 rad to 0.6588 rad (∼4.3%)** compared with the linear observer-based controller.

DiRT-ANN: Direct Radar Target Detection with Artificial Neural Networks *Sept 2024 – June 2025*

- Developed a **fully convolutional neural network (FCN)** to replace conventional CFAR detectors for a 60-GHz FMCW radar, trained on simulated range–azimuth maps and validated on real TI AWR6843ISK radar measurements.
- Achieved **99.25% detection accuracy** and **0.03 m range RMSE** ($R^2 = 99.59\%$), outperforming conventional CFAR detection across multiple SNR regimes.
- Built an end-to-end simulation, neural-network training, and hardware-validation pipeline using MATLAB, Python, TensorFlow, and AWR6843ISK. Project details and presentation

Deep Learning for Image Classification and Regression *Apr 2024 – June 2024*

- Implemented and evaluated **CNN, MLP, and Transformer architectures** for image classification and regression tasks on Stanford Dogs and FashionMNIST datasets.
- Developed the training and evaluation pipeline using **Python and PyTorch**, comparing model architectures and performance across learning tasks. Code on GitHub

ShapeSense: Real-Time Shape & Range Detector *Oct 2022 – Dec 2022*

- Built a real-time sensing system combining **computer-vision-based shape detection** with ultrasonic distance measurement and implemented the system on a BASYS3 FPGA board.
- Integrated Python/OpenCV vision processing with **VHDL-based digital hardware**, ultrasonic sensing, and real-time feedback. Code and video demonstration

Technical Skills

Programming: Python, C++, MATLAB, C, Embedded C, VHDL
Machine Learning & Scientific Computing: PyTorch, TensorFlow, JAX, NumPy, SciPy, CVXPY, OpenCV
Optimization & Probability: Linear/Convex Programming, LP Duality, Polyhedral Methods, Probabilistic Modeling
Formal Methods & Verification: Pacti, Z3, LTL/GR(1), SMT Solving, Quantifier Elimination, Contract Algebra
Embedded & Hardware: FPGA, VHDL, UART, DMA, Tiva C, BASYS3, Qt